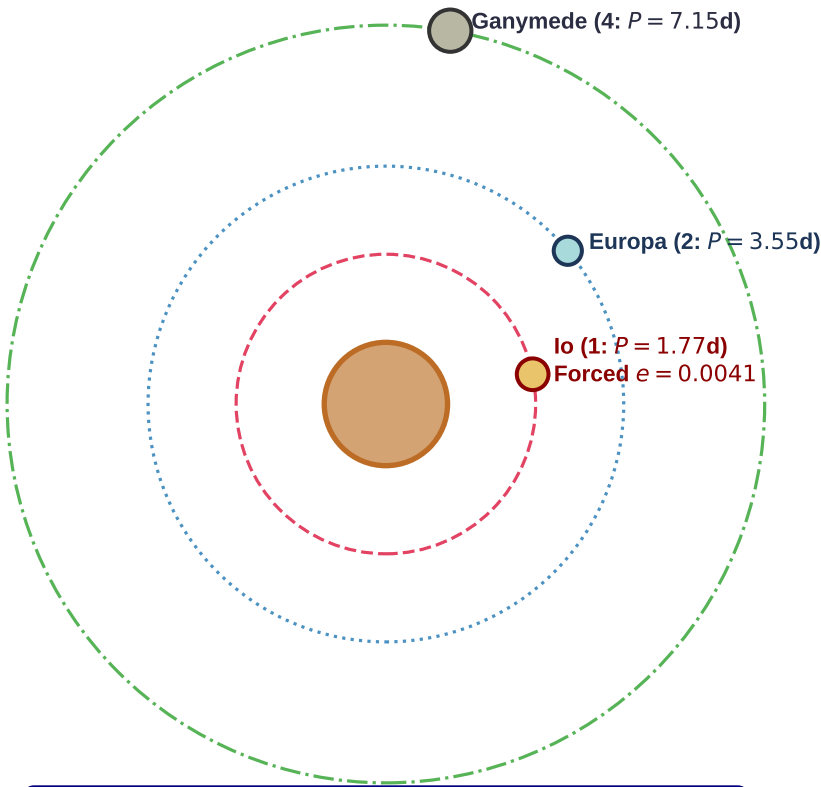
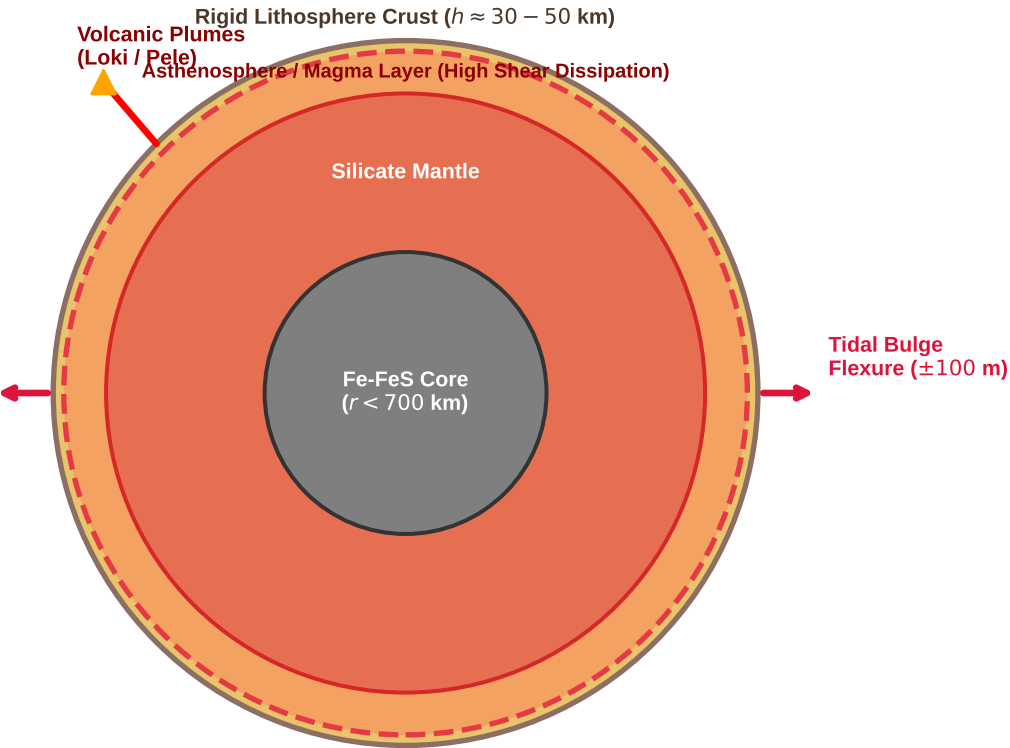


(a) Laplace Mean Motion Resonance Configuration



**4 : 2 : 1 Laplace Resonance :**  $n_{\text{Io}} - 3n_{\text{Eu}} + 2n_{\text{Ga}} = 0$   
Gravitational kicks continuously pump Io orbital eccentricity  $e_{\text{Io}}$ , preventing tidal circularization and sustaining continuous tidal heating.

(b) Io Interior Rheology & Cyclic Tidal Flexure



**Viscoelastic Dissipation :**  $\dot{\epsilon} = \frac{1}{2}\sigma_{ij}\dot{\epsilon}_{ij} = \frac{21}{2}\frac{k_2}{Q}\frac{GM_J^2R_{\text{Io}}^5n}{a^6}e^2$   
Cyclic shear deformation generates  $P_{\text{tide}} \approx 105 \text{ TW}$ , powering continuous volcanism.